

Internship Report

On

Ezlearn

D-9 Sector 3 Noida UP 201301



Session: Sem 8

Supervised by:

Mr Vishu P

(CTO at Ezlearn)

and

Dr Sunil Kumar

(Asst. Professor)

Submitted By:

Riyansh Garg

BTECH CSE

ENROLLMENT NUMBER. E22CSEU0062

**SCHOOL OF COMPUTER SCIENCE ENGINEERING AND
TECHNOLOGY, BU**

Certificate

This is to certify that the student named **Riyansh Garg (E22CSEU0062)** of Programme Name **BTECH CSE**, has undertaken an internship/industry project titled **“SDE Intern”** at **Ezlearn** from **12th January 2026** to **30th March 2026**. This progress report is a bona fide record of the work completed during the above period.

Date: 13th March 2026

Internal Guide / Faculty Mentor (Name & Signature)

Dr. Sunil Kumar

Industry Mentor (Name & Signature)

Mr. Vishu P

A handwritten signature in cursive script, reading "Vishu Pratap", written in black ink.

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Acknowledgement

I express my sincere gratitude to EzLearn for providing me the opportunity to undertake my Software Development Engineer (SDE) internship and to learn in a product-focused, fast-paced environment. The exposure to real-world engineering practices—spanning frontend, backend, DevOps, testing, and observability—has been invaluable for my professional growth. I am deeply thankful to my industry mentor, Mr. Vishu P (CTO, EzLearn), for his guidance, constructive feedback, and continuous encouragement throughout the internship. I would also like to thank the Engineering Team for their support in code reviews, technical discussions, and collaborative problem solving, which helped me build both technical and teamwork skills. I extend my appreciation to my Department and Academic Supervisor at Bennett University, whose mentorship and administrative support enabled me to align my internship objectives with academic requirements. Finally, I would like to thank everyone at EzLearn for the welcoming and collaborative culture that made this internship a rich learning experience

Offer Letter

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INTERNSHIP OFFER LETTER

To:
Riyansh Garg
Delhi

Mon, 12 Jan, 2025

Dear Riyansh Garg,

We are pleased to offer Riyansh Garg the position of Software Development Engineer Intern at Ezprep Company, with a Pre-Placement Offer (PPO) for the role of Software Development Engineer – SDE 1, subject to successful completion of the internship.

Internship Details

Candidate Name: Riyansh Garg
Position: Software Development Engineer Intern
Internship Duration: As per mutual agreement
Work Location: Remote / Hybrid (as per company policy)
Stipend: ₹25,000 per month

Pre-Placement Offer (PPO) Details:

Upon successful completion of the internship and performance evaluation, we are pleased to extend a Pre-Placement Offer for the role of Software Development Engineer – SDE 1 with the following terms:

Position: Software Development Engineer (SDE 1)
Employment Type: Full-time
Annual CTC: ₹9,00,000 per annum (9 LPA)

Responsibilities (Internship + Full-time)

- Develop and maintain application features and core modules
- Write clean, efficient, and well-documented code
- Work with modern frameworks, APIs, and databases
- Collaborate with senior engineers and product teams
- Participate in code reviews and technical discussions

We believe Riyansh's technical skills, learning ability, and dedication make him a strong fit for our engineering team, and we look forward to his continued growth with Ezprep.

Sincerely,
Ezprep

Background

EzLearn is an India-focused ed-tech platform that provides affordable, practice-driven engineering courses under a single subscription. The product vision is to help students progress from fundamentals (Aptitude, DSA, OS, DBMS, CN) to job-ready skills (Web, Cloud, DevOps, Data/AI) with **AI-generated personalized** roadmaps, hands-on projects, and a built-in **job portal**.

To support this vision at scale, EzLearn is building a modern web platform with:

- A responsive learner app (catalog, playlists, progress, quizzes, projects, certificates)
- A teacher/admin console (content ingestion, module/version control, review workflows)
- Subscription & payments, analytics, and growth funnels
- Robust infrastructure for media delivery, search, monitoring, and CI/CD

The SDE Internship was created to accelerate delivery on these tracks while giving the intern end-to-end exposure to product engineering in a startup context—requirements to release.

Organization: EzLearn (EdTech)

Intern Role: Software Development Engineer (SDE)

Internship Mode: Remote

Duration: 12th January 2026 to 30th March 2026

Purpose of Internship

The purpose of this report is to document the intern's role, objectives, methods, and outcomes within EzLearn's engineering roadmap. Specifically, the report aims to:

1. Describe how the intern contributed to feature development (frontend + backend) aligned with product milestones.
2. Capture engineering practices adopted (Git workflows, code reviews, testing, CI/CD, observability).
3. Summarize technologies and tools used to deliver reliable, maintainable, and performant increments.
4. Define the scope of work, constraints, and success criteria that guided the internship.
5. Reflect on learning outcomes—technical depth, product thinking, collaboration, and ownership.

Technology and Methods Used

Frontend

- **React (Next.js)** with **TypeScript** for application pages, App Router, and server components where applicable.
- **Tailwind CSS** with reusable component patterns to maintain a consistent UI.
- **Form management** using **React Hook Form** with **Zod** for validation.
- **State management** using **React Query** or **Server Actions** for efficient data fetching and caching.
- **Accessibility** practices including semantic HTML, keyboard navigation, and basic **WCAG guidelines**.
- **Media playback** support with **HLS/MP4 player integration**, chapter markers, and resume progress functionality.

Backend & APIs

- **FastAPI (Python)** or **Node.js with Express** (depending on the implementation used) for building REST APIs.
- **PostgreSQL** as the primary relational database for managing users, courses, modules, progress, and payments.
- **Prisma** or **SQLAlchemy** for database schema management and migrations.
- **Redis (optional)** for caching frequently accessed data and improving performance.
- **Authentication & Authorization** using **JWT-based sessions** or **NextAuth**, with **Role-Based Access Control (RBAC)** for roles such as admin, teacher, and learner.
- **Search functionality** using **PostgreSQL full-text search** or **OpenSearch / Meilisearch** for course discovery.
- **File and media storage** using **object storage services** (e.g., **Amazon S3 / Cloudflare R2**) with **signed URLs** and a **CDN** for optimized delivery.

Scope of Internship

1.4.1 Objectives

- Deliver production-grade increments across at least two product areas, such as:
 1. **Authentication & Onboarding** (signup/login, email OTP, RBAC, profile)
 2. **Course Catalog & Search** (filters, tags, relevance)
 3. **Player & Progress** (resume points, module completion, quizzes)
 4. **Admin Console** (course CRUD, module versioning, content QA)
 5. **Subscriptions & Billing** (plan gating, payments, webhooks)
- Maintain a high bar for code quality, tests, and documentation.
- Participate in grooming, standups, code reviews, and release planning.

1.4.2 Responsibilities

- Translate product specs/UX into technical tasks; estimate, implement, and demo.
- Write unit/integration tests; fix defects arising from QA or production logs.
- Improve performance with caching, pagination, and query optimization.
- Contribute to CI/CD workflows and deployment checklists.

1.4.3 Deliverables

- Merged PRs for scoped features (linked issues & commit history).
- Test coverage artifacts and CI run logs.
- Demo notes or short Looms for stakeholder review.
- A final retrospective: what shipped, impact, and next steps.

1.4.4 Success Metrics (Examples)

- Feature completion vs. sprint plan ($\geq 90\%$ on committed scope).
- Page/API performance improvements (e.g., p95 latency ↓, LCP < 2.5s on 75th pct).
- Support tickets for the touched areas trending down post-release.
- Positive reviewer comments and reduced cycle time for subsequent PRs.

1.4.5 Constraints & Out-of-Scope

- Payment gateway onboarding and legal compliance beyond engineering integration.
- Content production/editing — handled by teaching/content teams.
- 3rd-party contract negotiations or vendor management.

About Ezlearn

EzLearn is an India-focused ed-tech company delivering affordable, practice-driven engineering education through a single subscription. The platform combines structured video courses, hands-on projects, quizzes, and AI-generated personalized roadmaps to help learners move from fundamentals (Aptitude, DSA, OS, DBMS, CN) to job-ready skills (Web, Cloud, DevOps, Data/AI). A built-in job portal and resume builder bridge coursework to employability.

- **Business Model:**

Subscription (all-access), with select entry courses offered à-la-carte for trial/lead generation.

- **Delivery:**

Web app (responsive), with an admin/teacher console for content ingestion, review, and versioning.

- **Engineering Approach:**

Modern full-stack (React/Next.js + TypeScript; Node/FastAPI; PostgreSQL; Docker; CI/CD), data-informed product iteration, and lean, experiment-driven growth.

Vision:

Make high-quality engineering education accessible and outcome-oriented for students across Tier-2/3 cities and beyond.

Hierarchical Structure & Relevant Office Details

EzLearn follows a modern and collaborative organizational structure designed to support product engineering, customer learning, platform operations, and analytics. The company is led by its CTO, Mr Vishu P, followed by heads of major departments including Engineering, Product, Design, Sales, Marketing, Customer Success, and Operations. Each department consists of managers, team leads, and cross-functional product teams that work together to deliver high-quality educational solutions. EzLearn operates in a Hybrid model, providing flexibility for employees to choose their work environment and schedule. The company follows flexible working hours to maintain a healthy balance between productivity and personal well-being. Within the Engineering team, work is project-driven with clear divisions across frontend, backend, QA, DevOps, and product functions. The company follows an Agile development process, conducting two-week sprints with sprint planning and review meetings to ensure cross-team collaboration, transparency, and alignment on ongoing work and future goals. Communication and collaboration are managed through modern tools such as Slack, GitHub, and Jira, which facilitate real-time discussions, stand-ups, task tracking, and code reviews. This structured workflow encourages teamwork, autonomy, and continuous improvement, fostering a highly productive and collaborative culture at EzLearn.

Duration and Department Assigned

During the internship, work and responsibilities were distributed across multiple functional areas to provide broad exposure to the engineering ecosystem. The assigned departments and functions included:

- Product Engineering (Frontend & Backend) Involvement in building and enhancing user-facing features, backend integrations, and overall product functionality.
- Platform & DevOps Exposure to deployment workflows, CI/CD pipelines, and platform-level tooling essential for scalable product delivery.
- QA & Testing Participation in QA cycles, debugging, test-case validation, and ensuring product reliability before releases.
- Data & Analytics / Telemetry Working with product telemetry, analytics instrumentation, and understanding data-driven decisionmaking.
- Support & Release Operations (limited, as required) Assisting in triage, hot-fixes, and understanding the release processes when needed.

Major Activities

During the internship, I contributed to the development and improvement of the EzLearn learning platform through both frontend and backend engineering tasks.

On the frontend, I built and enhanced user-facing pages such as the course catalog, lesson player, and profile using Next.js with TypeScript. I implemented authentication flows including login, signup, OTP verification, and role-based access control. I also developed search and filtering features for the catalog and integrated an HLS-based video player with resume progress tracking and analytics events.

On the backend, I developed REST APIs using Node.js and Express for courses, modules, user progress, and quiz submissions. I worked with PostgreSQL for database design, wrote migrations, and optimized queries using indexing and pagination.

I also contributed to DevOps and quality assurance activities by containerizing services with Docker, supporting CI/CD pipelines using GitHub Actions, writing unit and integration tests, and assisting in debugging and resolving production issues.

Through these activities, I gained practical experience in full-stack development, product engineering workflows, and building scalable web applications in an Agile environment.

Skills Gained from the Role

During the internship at EzLearn, I developed both technical and professional skills while working in a real-world product engineering environment.

- **Full-Stack Development Skills:**

I strengthened my ability to build scalable web applications using modern technologies such as Next.js, React, TypeScript, Node.js, and PostgreSQL. I gained experience designing REST APIs, managing database schemas, and implementing efficient data-fetching patterns and caching strategies.

- **DevOps & Deployment Practices:**

I learned how production systems are built and deployed using Docker and CI/CD pipelines with GitHub Actions. I also gained exposure to environment configuration, deployment workflows, and maintaining reliable staging and production releases.

- **Testing & Quality Assurance:**

I developed skills in writing unit, integration, and API tests using tools like Jest and PyTest. I also gained experience in performance profiling, regression testing, and validating features through post-deployment smoke tests.

- **Observability & Data Awareness:**

I learned how to instrument analytics events, implement structured logging, and monitor system health using basic dashboards and uptime checks to understand product performance and user behavior.

- **Collaboration & Professional Skills:**

Working in an Agile team environment helped me improve communication, task estimation, code reviews, and documentation practices. I also developed problem-solving and debugging skills while handling real production issues and feature enhancements.

Conclusion: Overall Experience and Personal Takeaways

Overall, my internship at EzLearn was a highly valuable learning experience that allowed me to gain practical exposure to real-world product development. I had the opportunity to work across the full stack, contributing to frontend interfaces, backend APIs, database design, and deployment workflows while collaborating with a professional engineering team.

This experience helped me understand how modern web platforms are designed, built, tested, and deployed in a structured development environment. I also developed stronger problem-solving skills, improved my ability to write clean and maintainable code, and learned how to work effectively within Agile development processes.

Beyond technical knowledge, the internship strengthened my communication, collaboration, and documentation skills while working on real product features and resolving issues in a production codebase. Overall, the experience significantly improved my confidence as a software developer and provided a strong foundation for my future career in software engineering.

General Description of Tasks Done

During the reporting period, I contributed across the full stack of EzLearn's web platform, focusing on improving reliability, performance, and overall developer efficiency. My work spanned frontend development, backend APIs, database design, DevOps pipelines, observability, and QA processes. Below is a structured breakdown of the major (non-proprietary) tasks and projects completed.

A) Frontend Development (Next.js + TypeScript)

I implemented and enhanced several user-facing experiences using **Next.js (TypeScript)** and modern UI patterns.

- Built authentication flows including **login, signup, and session handling**.
- Implemented **catalog filters** (tags, level, duration) with **debounced search** and **URL-synced state** for consistent navigation.
- Enhanced the **lesson player** by managing states such as **resume position, loading sequences, and completion tracking**.
- Added **client-side validations** using **React Hook Form and Zod**, improving input reliability and preventing invalid submissions.

B) Backend API Development (Node.js / Express)

I designed and refined several **REST API endpoints** powering core platform functionality.

- Created APIs for **courses, modules, user progress, and quiz submissions**, enabling end-to-end learning flows.
- Added **request validation and structured error responses** for predictable API behavior.
- Optimized routes with **pagination and selective field returns** to reduce payload sizes and improve performance.

C) Database Design & Optimization (PostgreSQL)

I contributed to **data modeling and query optimization** across high-traffic entities.

- Designed **schema tables** for content, progress, and metadata.
- Wrote **database migrations** for structural updates.

- Added **indexes on frequently queried fields** and introduced **pagination strategies** to improve query performance.

D) CI/CD Pipeline Enhancements (GitHub Actions)

I helped configure and improve EzLearn's **automated deployment workflows**.

- Set up **CI pipelines** covering build, lint, type-check, and testing stages.
- Added **deployment guardrails** so releases proceed only after passing all required checks.
- Supported **staging** → **production workflows** using branch protections and automated deployment steps.

E) Observability & Monitoring Improvements

I introduced **baseline observability and system health monitoring**.

- Added **structured logs** for API tracking and debugging.
- Implemented **uptime checks and health probes** for service reliability.
- Contributed to **small dashboards** displaying API performance and error trends.

F) Testing & Quality Assurance

I actively contributed to the **testing lifecycle across multiple layers**.

- Wrote **unit tests and integration tests** for critical application flows.
- Set up **smoke tests** to validate deployments after CI/CD rollouts.
- Helped **debug regressions** reported by QA teams or surfaced during CI runs.

These efforts helped **improve platform stability, maintain code quality, and reduce post-deployment failures**.

Milestones Achieved

During the internship, several important milestones were achieved that contributed to the development, stability, and performance of the EzLearn learning platform. These milestones reflect the successful delivery of key product features and engineering improvements.

A) Authentication & Onboarding Implementation

- Successfully implemented **Email/OTP-based authentication**, password reset functionality, and **role-based access control (RBAC)**.
- Deployed the complete authentication flow from **staging to production** with stability checks and validation.

B) Course Search & Filter System

- Developed **debounced search and URL-based filters** for course discovery, including filters for topic, difficulty level, and duration.
- Improved the **user experience and navigation consistency** across the course catalog.

C) Lesson Player Progress Tracking

- Implemented **resume points and completion tracking** within the lesson player.
- Enabled **progress synchronization across sessions and devices**, ensuring continuity for learners.

D) API Performance Optimization

- Improved backend performance by introducing **database indexing and pagination** on high-traffic endpoints.
- Reduced latency in course listing APIs and improved response efficiency.

E) CI/CD Pipeline Improvements

- Established **quality gates** in CI/CD pipelines by enforcing linting, type-checking, and automated tests before merges.
- Implemented **post-deployment smoke tests** and documented rollback procedures to ensure reliable deployments.

F) Telemetry & Analytics Integration

- Instrumented key product analytics events such as **user onboarding steps, search activity, video playback, and quiz submissions**.
- Enabled **early insights into user behavior and product engagement** through basic analytics dashboards.

These milestones helped strengthen the platform's reliability, improve performance, and enhance the overall learning experience for users.

Skills / Tools Learned

During the internship, I gained hands-on experience with a variety of modern tools, technologies, and development practices used in real-world software engineering environments.

Programming Languages & Frameworks

- **Next.js (TypeScript)** for building scalable frontend applications.
- **React**, along with **React Hook Form** and **Zod**, for form handling and validation.
- **Node.js and Express** for developing REST APIs and backend services.
- **PostgreSQL** for relational database design and query optimization.
- **Docker** for containerizing services and managing application environments.

Development & Collaboration Tools

- **Git and GitHub** for version control, branching strategies, and pull request workflows.
- **GitHub Actions** for implementing CI/CD pipelines including build, test, and deployment automation.
- **Postman and OpenAPI** for API testing, debugging, and documentation.

Testing & Quality Assurance

- Conducted **smoke testing and regression validation** after deployments.
- Performed **Core Web Vitals checks and API query profiling** to monitor performance.

Analytics & Telemetry

- Implemented **custom event instrumentation** for tracking user interactions such as onboarding, search, video playback, and quiz submissions.
- Built **basic funnel analysis dashboards** to understand user behavior and engagement.
- Used **feature flags** for controlled rollout of new features.

Methods & Development Processes

- Worked within an **Agile development environment** including sprints, stand-ups, sprint planning, and retrospectives.
- Participated in **code reviews and collaborative development practices**.
- Followed **versioned releases with tagged builds**, along with documentation of **runbooks and rollback workflows** for reliable deployments.

Reflections on Challenges Faced and Contributions Made

During the internship, I encountered several real-world engineering challenges while working on a live production platform. Addressing these challenges helped me strengthen my debugging, system design, and problem-solving skills while contributing to improvements in the platform's reliability and performance.

1. Maintaining State Consistency Between Client and Server

Challenge:

When users switched lessons quickly or experienced network interruptions, progress updates sometimes conflicted or failed to synchronize correctly between the client and server.

Solution / Contribution:

I implemented **idempotent update endpoints**, introduced **optimistic UI updates**, and added **rollback handling** to ensure that client and server states remained consistent even during rapid user interactions.

2. Slow Search Performance with Multiple Filters

Challenge:

Applying multiple filters such as topic, difficulty, duration, and tags caused slower database queries and delayed responses in the course catalog.

Solution / Contribution:

I improved performance by adding **composite database indexes**, optimizing **query execution plans**, and implementing **server-side pagination**. Additionally, non-critical fields were moved off the main query path to reduce load and improve response times.

3. Unreliable Deployments and Regression Issues

Challenge:

Early deployments occasionally introduced regressions due to limited automated checks and validation steps in the deployment pipeline.

Solution / Contribution:

I helped strengthen the **CI/CD pipeline** by enforcing **linting**, **type checking**, and **automated test gates** before

merges. I also added **post-deployment smoke tests** and documented a **rollback procedure** to ensure safer and more reliable deployments.

4. Validation Mismatch Between Client and Server

Challenge:

Differences between frontend and backend validation rules caused inconsistent or unclear error messages for users.

Solution / Contribution:

To address this, I centralized **validation schemas**, standardized **backend error structures**, and reused the same validation rules across both the client and server, ensuring consistent input validation and clearer feedback to users.

Through these challenges, I contributed to improving the **platform's stability, performance, and development workflows**, while gaining valuable experience working with real production systems and collaborative engineering environments.

Weekly Work Logs

Week No.	Dates (From–To)	Task Description	Hours Spent	Mentor Signature
1	Feb 1 - Feb 7	Worked on improving EzLearn's authentication and onboarding flow by implementing login and signup with Email/OTP verification and role-based access control. Added client-side validation using React Hook Form and Zod to ensure reliable input handling.	40	<i>Vishu Pratap</i>
2	Feb 8 - Feb 14	Implemented catalog search and filtering features for the course discovery page. Added debounced search queries and filters for topic, difficulty level, and duration. Introduced URL-synced filter state so that catalog results remained consistent during navigation and page reloads.	40	<i>Vishu Pratap</i>
3	Feb 15 - Feb 21	Developed backend REST APIs using Node.js and Express for core entities including courses, modules, user progress, and quiz submissions. Implemented structured request validation and standardized error responses to improve API reliability and maintainability.	40	<i>Vishu Pratap</i>
4	Feb 22 - Feb 28	Enhanced the lesson player by implementing resume progress tracking and completion state updates. Ensured that user progress remained synchronized between the client and server during rapid navigation or unstable network conditions.	40	<i>Vishu Pratap</i>
5	March 1 - March 7	Worked on PostgreSQL database design by creating schemas for courses, modules, and progress tracking. Added indexes and pagination to optimize query performance for frequently accessed endpoints.	40	<i>Vishu Pratap</i>
6	March 8 - March 14	Developed REST APIs for courses, modules, user progress, and quiz submissions using Node.js and Express. Implemented request validation and structured error responses to improve API reliability.	40	<i>Vishu Pratap</i>
7	March 15 - March 16	Improved deployment reliability by contributing to GitHub Actions CI/CD pipelines, adding build, lint, and testing checks before deployments.	16	<i>Vishu Pratap</i>

Annexures

Approval for College Documentation Reports External Inbox x

◆ Summarise this email



Vishu Pratap <vishup@ezlearn.in>

to Tanishk, me, khushig ▾

Hi,

This is to confirm that your weekly logs have been approved. You are authorized to use them for your college report. My signature is attached; kindly use it only in the agreed-upon locations.

Regards

Vishu P

CTO

EzLearn

One attachment • Scanned by Gmail ⓘ

Mentor can be contacted at vishup@ezlearn.in or 7982114673